Neel Gali

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of hands-on experience in building predictive models and advanced ML models, driving continuous improvement pipelines from research to production. Proficient in training and fine-tuning models, including open-source models, utilizing strong Python development skills with PyTorch and TensorFlow to deliver scalable solutions.

TECHNICAL SKILLS

Programming & Core ML: Python development skills, SQL, Scikit-learn, NumPy, Pandas, predictive modeling, feature engineering, model evaluation

Deep Learning & AI: PyTorch, TensorFlow, Keras, transformer architectures, deep learning, reinforcement learning, computer vision, natural language processing, open-source models, advanced ML models, training and fine-tuning models, develop algorithms from scratch

Cloud & MLOps: AWS, Azure, GCP, Docker, Kubernetes, MLflow, CI/CD, Git, MLOps, continuous improvement pipelines

Databases & Data Management: PostgreSQL, MySQL, MongoDB, Apache Kafka, data warehousing

PROFESSIONAL EXPERIENCE

AI / ML Engineer | Epik Solutions

Jun 2025 - Present

• Designed and implemented robust predictive models using Python development skills and TensorFlow to forecast cybersecurity threats; achieved 92% accuracy in anomaly detection, enhancing threat intelligence for internal security teams and reducing false positives by 40%.

• Led the training and fine-tuning models process for transformer architectures applied to network telemetry data, improving real-time intrusion detection capabilities by 30% for critical infrastructure protection, used by incident response teams.

• Developed continuous improvement pipelines for machine learning models, integrating MLOps best practices with Docker and Kubernetes on AWS, reducing model deployment time by 50% and ensuring rapid adaptation to evolving threat landscapes.

• Engineered novel advanced ML models from scratch using PyTorch for zero-day exploit prediction, which detected previously unknown threats with 85% precision, significantly strengthening organizational security posture.

• Evaluated and optimized open-source models for threat classification, customizing architectures to meet specific enterprise security requirements; resulted in a 25% reduction in manual alert investigations for the SOC team.

• Collaborated with cross-functional teams to translate complex security challenges into quantifiable deep learning objectives, delivering solutions that improved operational efficiency by 20% and supported compliance audits.

AI / ML Engineer | Samay Systems LLC

July 2024 - Jun 2025

• Built and deployed predictive models using Python and Scikit-learn to identify compliance risks across cloud environments, decreasing audit preparation time by 35% for legal and compliance teams.

• Contributed to continuous improvement pipelines for machine learning workflows, leveraging Azure DevOps for automated model validation and retraining, ensuring 99.9% data integrity in compliance evidence collection.

• Implemented training and fine-tuning models strategies for open-source models to process unstructured compliance documents, achieving 90% accuracy in automated clause extraction and reducing manual review efforts by 50%.

• Designed and developed advanced ML models for fraud detection in financial compliance, utilizing TensorFlow on Azure to process large datasets, leading to a 15% reduction in potential financial losses.

• Leveraged transformer architectures in natural language processing models to summarize complex regulatory documents, improving the efficiency of compliance analysts by 25% and ensuring quicker decision-making.

• Architected scalable data ingestion pipelines for multi-cloud compliance data, ensuring high-quality inputs for deep learning models and supporting robust, auditable data governance frameworks.

ML Intern | Monarch Tractor

Jun 2023 - Aug 2023

• Assisted in the training and fine-tuning models for object detection using PyTorch and OpenCV for autonomous agricultural vehicles, achieving 95% accuracy in crop and obstacle identification in diverse field conditions.

• Developed Python scripts for data preprocessing and augmentation for deep learning models, generating synthetic datasets that expanded training data by 300% and improved model robustness.

• Contributed to continuous improvement pipelines for model evaluation by automating metric calculation and visualization, reducing iteration cycles by 20% and accelerating development of predictive models.

• Researched and implemented various open-source models for semantic segmentation, evaluating their performance against real-world agricultural scenarios to inform the selection of advanced ML models for production.

• Gained hands-on experience deploying machine learning models to edge devices, optimizing TensorFlow Lite models for real-time inference with minimal latency on vehicle hardware.

• Collaborated closely with senior engineers to validate model outputs against operational constraints, ensuring safety and efficiency for semi-autonomous systems and contributing to improved system reliability.

Associate Data Scientist | ZeMoSo Technologies

Jul 2021 - Aug 2022

• Supported the development of predictive models using Python and Scikit-learn to optimize laboratory operations, reducing experiment run times by 10% and improving resource allocation.

• Performed extensive data preprocessing and feature engineering for diverse experimental datasets, ensuring high-quality inputs for machine learning models and improving downstream analysis accuracy by 15%.

• Assisted in setting up foundational continuous improvement pipelines for data validation and model versioning, using Git and Jenkins to ensure reproducibility and auditability of scientific workflows.

• Contributed to the exploration of deep learning applications for image-based experiment analysis, researching open-source models to enhance data extraction efficiency for laboratory automation platform.

EDUCATION

Master’s in Computer Science | Oklahoma State University, Stillwater, OK

Bachelor’s in Computer Science | JNTU Hyderabad, India